ALICE L WALTON SCHOOL OF MEDICINE

Treatment of Dyslipidemia

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Learning Objectives (3)

After completing this brick, you will be able to:

1. Explain indications for LDL-C and triglyceride lowering therapy based on patient risk factors
2. Apply treatment guidelines and ASCVD risk assessment to select appropriate lipid-lowering therapies
3. Evaluate patient scenarios to optimize and adjust dyslipidemia treatment strategies

CASE CONNECTION

ML, a 56-year-old man, presents at his primary care clinic with elevated cholesterol (245 mg/dL), LDL-C (162 mg/dL), HDL (47 mg/dL), and triglyceride levels (180 mg/dL) discovered during routine labs. ML has a history of hypertension (diagnosed 10 years ago) and type 2 diabetes (diagnosed 5 years ago). His BP=155/84 mmHg, Hgb A1C=6.7%, BMI= 36, and eGFR=59. He quit smoking 2 years ago after a 25-pack-year history. He takes lisinopril and metformin. He does not currently take any lipid-lowering medication. His father had an MI at age 52.

You are a second-year medical student working in the clinic, tasked with reviewing lipid profiles and recommending treatment approaches.

How do you decide on initial therapy to address his elevated LDL-C and triglycerides? What treatment would you recommend for ML?

Consider your answers as you read, and we'll revisit ML at the end of the brick.

GO TO CONCLUSION ↓

What Are the Primary Goals of Dyslipidemia Treatment Focusing on LDL-C and Triglyceride Lowering? (LO#1)

Dyslipidemia treatment aims to reduce the risk of atherosclerotic cardiovascular disease (ASCVD) by targeting abnormal lipid levels, primarily elevated low-density lipoprotein cholesterol (LDL-C) and triglycerides (TG). Extensive clinical evidence demonstrates that lowering LDL-C significantly reduces cardiovascul [illegible] glycerides provides additional benefit in specific patient populations. Treatment goals are individualized based on a patient's overall cardiovascular risk profile, with more aggressive targets for those at higher risk.

Cardiovascular Risk Reduction

The fundamental goal of dyslipidemia treatment is to reduce the risk of clinical ASCVD events, including myocardial infarction, stroke, and peripheral arterial disease. Elevated LDL-C plays a causal role in atherosclerosis development through its contribution to arterial plaque formation. Multiple randomized controlled trials and meta-analyses have consistently demonstrated that LDL-C reduction, particularly with statin therapy, leads to proportional decreases in cardiovascular events, with approximately **22% reduction in major vascular events for every 40 mg/dL (1 mmol/L) decrease in LDL-C**.

Elevated triglycerides also contribute to cardiovascular risk, though the relationship is more complex. Hypertriglyceridemia promotes atherosclerosis through several mechanisms: increased levels of remnant lipoproteins, enhanced formation of small dense LDL particles, and reduced high-density lipoprotein cholesterol (HDL-C) levels. Recent evidence from genetic studies and clinical trials with triglyceride-lowering agents suggests that reducing triglycerides, particularly in patients with elevated levels (≥150 mg/dL), can provide additional cardiovascular benefit beyond LDL-C reduction alone.

LDL-C and Triglyceride Targets

Current guidelines from major cardiovascular societies, including the American College of Cardiology/American Heart Association (ACC/AHA) and the European Society of [illegible] Society (ESC/EAS), recommend specific targets for lipid parameters based on individual risk assessment. For LDL-C, the target values are increasingly aggressive for higher-risk patients, reflecting the "lower is better" principle established through clinical trials.

The following table outlines the general LDL-C targets based on risk categories:

Table

Risk Category	LDL-C Target	Percent Reduction
Very high risk (secondary prevention, diabetes with target organ damage)	<55 mg/dL (<1.4 mmol/L)	≥50% from baseline
High risk (multiple major risk factors, diabetes without target organ damage)	<70 mg/dL (<1.8 mmol/L)	≥50% from baseline
Moderate risk (younger patients with one risk factor)	<100 mg/dL (<2.6 mmol/L)	30-50% from baseline
Low risk (no significant risk factors)	<116 mg/dL (<3.0 mmol/L)	Consider if >116 mg/dL

For triglycerides, there are no specific target values in most guidelines, but levels are categorized as follows:

- Normal: <150 mg/dL (<1.7 mmol/L)
- Borderline high: 150-199 mg/dL (1.7-2.3 mmol/L)
- High: 200-499 mg/dL (2.3-5.6 mmol/L)
- Very high: ≥500 [illegible]

Triglyceride levels ≥500 mg/dL require immediate intervention primarily to prevent pancreatitis, while levels between 150-499 mg/dL are addressed to reduce cardiovascular risk, particularly in patients with other risk factors. The primary focus remains on LDL-C reduction, with triglyceride management as a secondary target, especially in patients with diabetes, metabolic syndrome, or established ASCVD who have elevated triglycerides despite statin therapy.

What are the main differences between LDL-C and triglyceride treatment goals?

What Are the Common Pharmacologic Agents Used to Lower LDL-C and Triglycerides and Their Mechanisms?

Multiple pharmacologic agents are available to manage dyslipidemia, each with distinct mechanisms of action and effects on lipid parameters. Understanding these mechanisms helps clinicians select appropriate therapies based on the specific lipid abnorm[illegible] have discussed the pharmacological and non-pharmacological treatment options earlier. Attempt the active learning exercise below to review your knowledge.

ACTIVE LEARNING QUESTION

Match the drug or drug class to its mechanism of action or Adverse drug reaction

Evolocumab	✓ Inhibition of PSCK9 activity
Ezetimibe	✓ Inhibition of Neiman Pick protein and cholesterol transport
Niacin	✓ Heptatoxicity
Simvastatin	✓ Contraindicated for pregnancy
Gemfibrozil	✓ PPAR-a agonist

Explanation

Evolocumab is a PSCK9 monoclonal antibody that inhibits the function of PSCK9 and increases the removal of cholesterol from the bloodstream. [illegible]port from the intestine by inhibiting the Neiman Pick protein. Niacin has a high risk for hepatotoxicity and is contraindicated in patients with severe hepatic disease. Simvastatin is contraindicated in pregnancy and can also cause myalgias. Gemfibrozil is PPAR-a activator that decreases TG levels and is often recommended for hypertriglyceridemia.

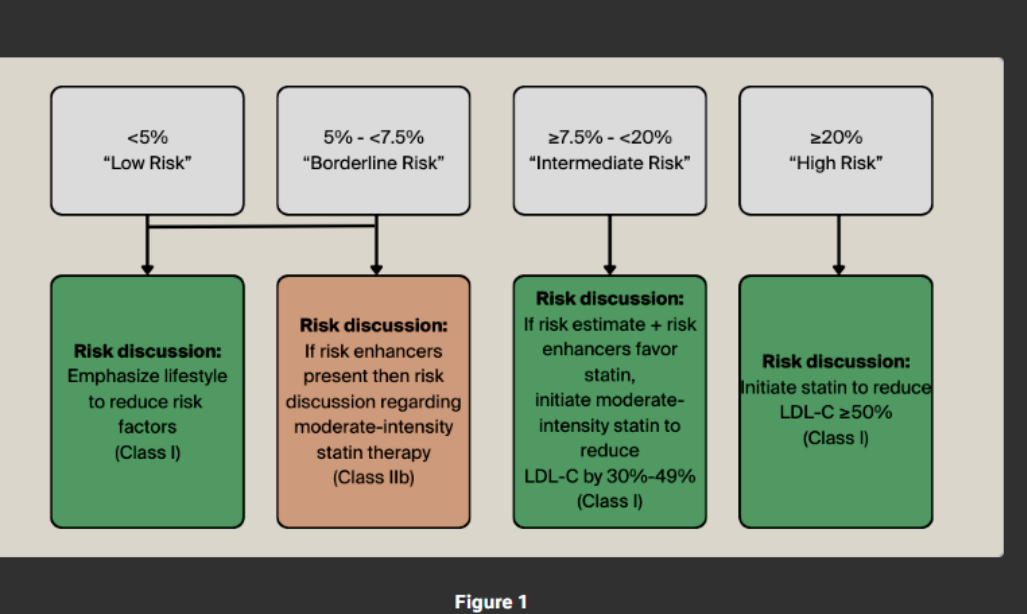
What Are the Indications for LDL-C and Triglyceride Lowering Therapy Based on Patient Risk Factors? (LO#2)

The decision to initiate lipid-lowering therapy depends on a comprehensive assessment of the patient's cardiovascular risk profile rather than isolated lipid values. Current guidelines emphasize a risk-based approach that considers multiple factors including age, sex, blood pressure, smoking status, diabetes, and lipid levels to determine the intensity and targets of therapy. This personalized approach ensures that those at highest risk receive appropriate treatment intensity while avoiding unnecessary medication in low-risk individuals.

Atherosclerotic Cardiovascular Disease (ASCVD) Risk Scoring

The cornerstone of determining treatment indications is ASCVD risk assessment. Patients are typically categorized into risk groups that guide therapy decisions (You can also follow along with Figure 1)

- Secondary prevention: Patients with established ASCVD (prior myocardial infarction, stroke, transient ischemic attack, stable or unstable angina, coronary or othe[illegible] arterial disease)
- Primary prevention with very high risk: Patients with multiple major risk factors, especially diabetes with target organ damage or familial hypercholesterolemia
- Primary prevention with high risk: Patients with significant risk factors or 10-year ASCVD risk score ≥7.5-20%
- Primary prevention with borderline-to-moderate risk: Patients with 10-year ASCVD risk score of 5-7.5%
- Primary prevention with low risk: Patients with 10-year ASCVD risk score <5%



<5% "Low Risk"
5% - <7.5% "Borderline Risk"
≥7.5% - <20% "Intermediate Risk"
≥20% "High Risk"
Risk discussion: Emphasize lifestyle to reduce risk factors (Class I)
Risk discussion: If risk enhancers present then risk discussion regarding moderate-intensity statin therapy (Class IIb)
Risk discussion: If risk estimate + risk enhancers favor statin, initiate moderate-intensity statin to reduce LDL-C by 30%-49% (Class I)
Risk discussion: Initiate statin to reduce LDL-C ≥50% (Class I)

Figure 1

The 10-year ASCVD-PREVENT calculator estimates a 10-year risk for CV disease, heart failure, and ASCVD for 30-79-year old. This "**ASCVD-PREVENT Calculator**" uses age, sex, systolic pressure, diabetes, smoking, blood pressure agents, [illegible] calculate the risk score.

Beyond traditional risk factors, certain conditions substantially increase ASCVD risk and may influence treatment decisions. **These risk enhancers include:**

- Family history of premature ASCVD (males <55 years, females <65 years)
- Primary hypercholesterolemia (LDL-C > 160 mg/dL)
- Metabolic syndrome
- Chronic kidney disease (eGFR 15-59 mL/min/1.73m²)
- Chronic inflammatory conditions (rheumatoid arthritis, psoriasis, HIV)
- History of premature menopause or preeclampsia
- High-risk ethnic groups (South Asian ancestry)
- Persistent elevations of triglycerides ≥175 mg/dL
- hs-CRP > 2.0 mg/L
- Lp(a) level > 125 mol/L
- apoB > 130 mg/dL
- Ankle-Brachial Index (ABI) < 0.9

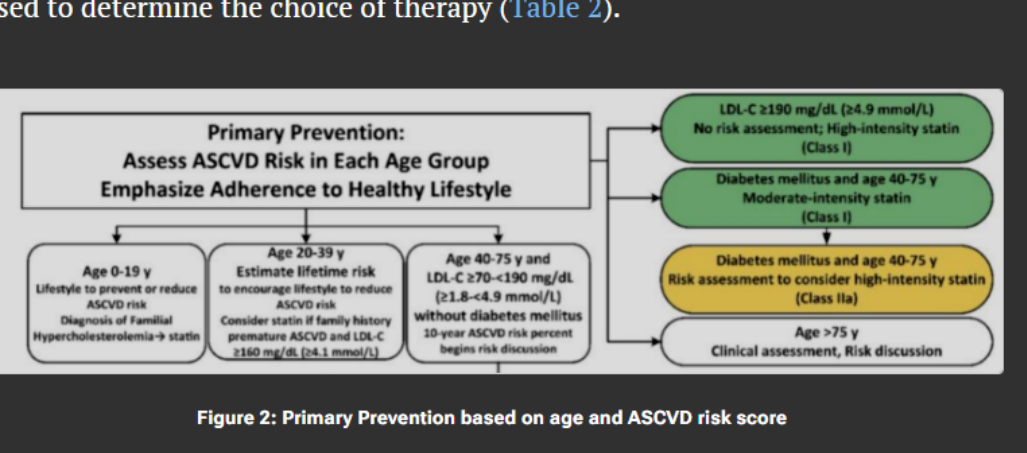
CLINICAL REASONING

Ankle-Brachial Index (ABI) < 0.9 doesn't measure cholesterol directly; it's a clinical tool to identify vascular disease burden that results from dyslipidemia. An ABI <0.5 can indicate severe arterial disease.

The presence of thes[illegible]**ensive therapy** or **earlier intervention**, particularly in patients with borderline or intermediate risk calculations.

Thresholds for Therapy Initiation

For LDL-C lowering therapy, current guidelines recommend the following thresholds for initiating pharmacologic treatment. Following the risk assessment with the ASCVD-PREVENT calculator, patient's are categorized as shown in Figure 2. A combination of factors (age, risk score, and diabetes) are used to determine the choice of therapy (Table 2).



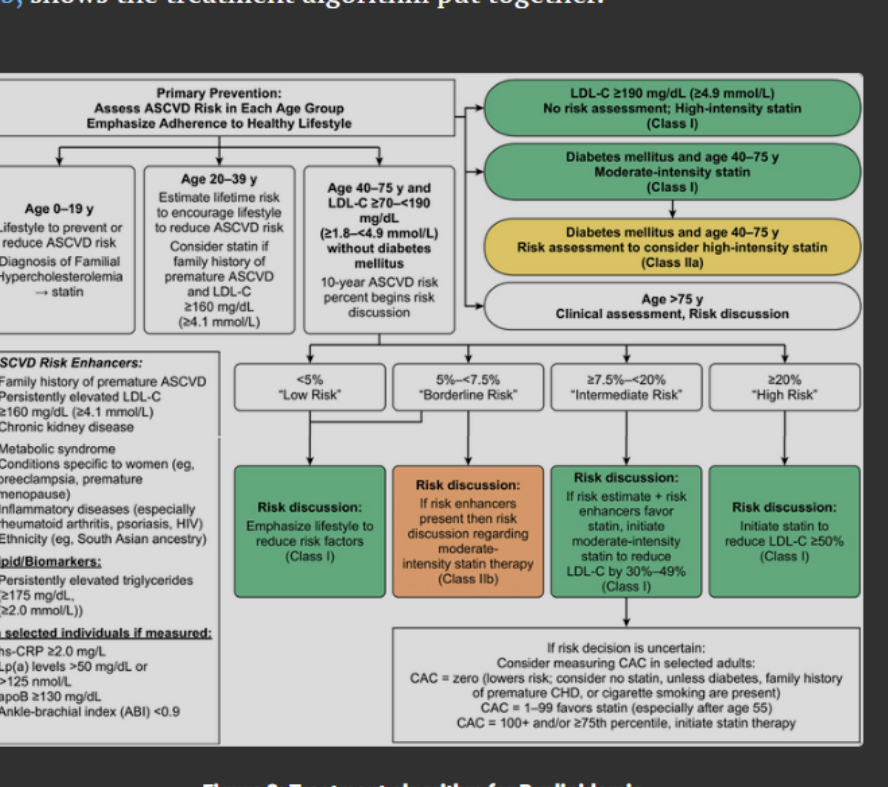
Primary Prevention:
Assess ASCVD Risk in Each Age Group
Emphasize Adherence to Healthy Lifestyle
Age 0-19 y
Age 20-39 y
Age 40-75 y and LDL-C ≥70-<190 mg/dL (≥1.8-<4.9 mmol/L) without diabetes mellitus
LDL-C ≥190 mg/dL (≥4.9 mmol/L) No risk assessment; High-intensity statin (Class I)
Diabetes mellitus and age 40-75 y Moderate-intensity statin (Class I)
Diabetes mellitus and age 40-75 y Risk assessment to consider high-intensity statin (Class IIa)
Age >75 y Clinical assessment, Risk discussion

Figure 2: Primary Prevention based on age and ASCVD risk score

Table 2

Patient Category	LDL-C Threshold for Considering Medication
Secondary prevention (established ASCVD)	Any LDL-C level (high-intensity statin recommended)
Primary prevention with LDL-C ≥190 mg/dL	≥190 mg/dL (high-intensity statin recommended)
Diabetes, age 40-75 years	[illegible]rate-intensity statin)
Primary prevention, age 40-75 years, 10-year risk ≥7.5%	≥70 mg/dL (moderate to high-intensity statin)
Primary prevention, age 40-75 years, 10-year risk 5-7.5%	≥70 mg/dL (consider moderate-intensity statin)
Primary prevention, age 40-75 years, 10-year risk <5%	≥190 mg/dL (consider moderate-intensity statin)

Figure 3, shows the treatment algorithm put together.



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LDL-C ≥190 mg/dL (≥4.9 mmol/L) No risk assessment; High-intensity statin (Class I)
Diabetes mellitus and age 40–75 y Moderate-intensity statin (Class I)
Diabetes mellitus and age 40–75 y Risk assessment to consider high-intensity statin (Class IIa)
Age >75 y Clinical assessment, Risk discussion
ASCVD Risk Enhancers:
Lipid/Biomarkers:
In selected individuals if measured:
<5% "Low Risk"
5%–<7.5% "Borderline Risk"
≥7.5%–<20% "Intermediate Risk"
≥20% "High Risk"
Risk discussion:
If risk decision is uncertain: Consider measuring CAC in selected adults

Figure 3: Treatment algorithm for Dyslipidemia

For triglyceride management, the thresholds for pharmacologic intervention are:

- **Severe** hypertriglyceridemia (≥500 mg/dL): Immediate pharmacologic intervention primarily to reduce pancreatitis risk
- **Moderate** hypertriglyceridemia (150-499 mg/dL): Consider pharmacologic therapy after lifestyle optimization in high-risk patients, particularly those with diabetes or established ASCVD
- **Mild** hypertriglyceridemia (150-199 mg/dL): Focus on lifestyle modifications and addressing secondary causes

CLINICAL REASONING

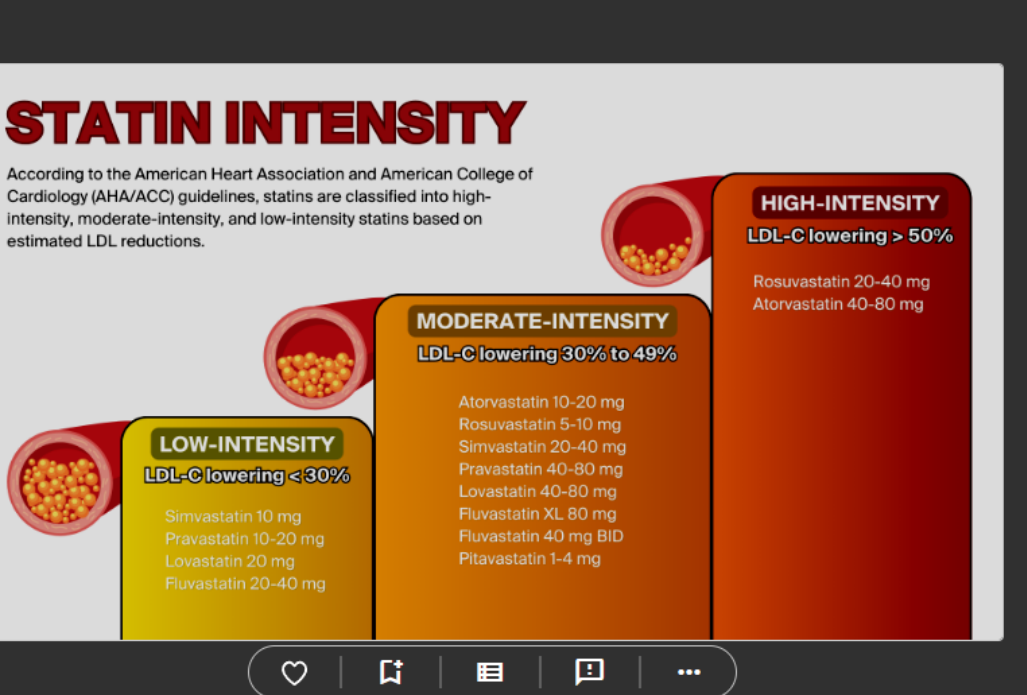
Secondary causes of hypertriglyceridemia should always be identified and addressed **before or concurrent with pharmacologic therapy**. These include uncontrolled diabetes, hypothyroidism, renal disease, pregnancy, and medications (estrogens, tamoxifen, glucocorticoids, retinoids, beta-blockers, thiazides, and antiretrovirals). Lifestyle factors such as excessive alcohol consumption, high carbohydrate intake, and obesity also significantly contribute to hypertriglyceridemia.

How Do We Apply Treatment Guidelines to Select Appropriate LDL-C and Triglyceride Lowering Therapies?

Applying treatment g[illegible]ach that considers the patient's risk profile, specific lipid abnormalities, comorbidities, and potential drug interactions. Current guidelines from major cardiovascular societies provide a framework for selecting appropriate therapies, but clinical judgment remains essential for individualizing treatment. The goal is to match the intensity of therapy to the level of risk while considering patient preferences and tolerability.

First-Line Therapy Selection

For LDL-C lowering, **statins remain the first-line therapy** for most patients due to their robust evidence base demonstrating both LDL-C reduction and cardiovascular outcome benefits. Statin intensity selection depends on the patient's risk category:



The recommended first-line therapy approach based on patient risk categories includes:

- **Secondary prevention** (established ASCVD): High-intensity statin to achieve ≥50% LDL-C reduction
- **Very high-risk primary prevention** (LDL-C ≥190 mg/dL, diabetes with target organ damage): High-intensity statin
- **High-risk primary prevention** (10-year risk ≥20%): Moderate to high-intensity statin
- **Intermediate-risk primary prevention** (10-year risk 5-7.5%): Moderate-intensity statin after risk discussion
- **Low-risk primary prevention** (10-year risk <5%): Lifestyle modification; consider statin only if LDL-C ≥190 mg/dL or strong family history

For triglyceride management, the approach depends on the severity of hypertriglyceridemia:

- **Severe** hypertriglyceridemia (≥500 mg/dL): Fibrates, prescription omega-3 fatty acids, or both as first-line therapy to reduce pancreatitis risk
- **Moderate** hypertriglyceridemia (150-499 mg/dL): Optimize statin therapy first, then consider adding icosapent ethyl (fish oil) in high-risk patients with triglycerides ≥150 mg/dL despite statin therapy
- **Mild** hypertriglyceridemia (150-199 mg/dL): Focus on lifestyle modifications and statin therapy if indicated based on overall risk

Special considerations for first-line therapy selection include:

- **Age:** Consider moderate-intensity statins in patients >75 years unless they have established ASCVD
- **Pregnancy pote**[illegible]sider bile acid sequestrants if treatment is necessary
- **Diabetes:** Statins may slightly increase diabetes risk but benefits outweigh risks; monitor glucose
- **Liver disease:** Use caution with statins in active liver disease; monitor liver function tests

Combination Therapy Considerations

When monotherapy with statins fails to achieve LDL-C goals or is not tolerated at higher doses, combination therapy becomes an important strategy. The sequential approach to combination therapy for LDL-C lowering typically follows:

1. **Maximize statin dose** to the highest tolerated intensity
2. **Add ezetimibe** if LDL-C remains above target despite maximally tolerated statin
3. **Add PCSK9 inhibitor** if LDL-C remains above target despite statin plus ezetimibe, particularly in very high-risk patient.

Important **drug interactions** to consider with combination therapy include:

- Gemfibrozil significantly increases statin exposure and myopathy risk; **avoid this combination**
- Cyclosporine, macrolide antibiotics, antifungals, and protease inhibitors (Paxlovid for COVID and combo treatments for HIV) can increase statin levels; dose adjustment may be needed

In case of **familial hypercholesterolemia**, high intensity statins remain as the first line treatment. If the LDL-C goals remain above target, PSCK-9 inhibitors and Ezetimibe can be used.

How Do We Evaluate and Adjust Dyslipidemia Treatment Strategies in Patient Scenarios? (LO#3)

Effective dyslipidemia management requires ongoing evaluation and adjustment of treatment strategies based on patient response, tolerance, and changing clinical circumstances. This dynamic approach ensures that therapy remains optimized for each patient's evolving risk profile and treatment goals. Regular monitoring allows clinicians to assess efficacy, detect adverse effects early, and make evidence-based adjustments to the treatment plan.

Monitoring and Follow-Up

After initiating lipid-lowering therapy, regular monitoring is essential to evaluate treatment efficacy and safety. The recommended monitoring schedule includes:

- Baseline **lipid panel and liver function tests** before starting therapy
- Repeat lipid panel 4-12 weeks after initiating therapy or dose changes to assess response
- Adherence to medication and lifestyle modifications

CLINICAL CORRELATION

Liver function tests 4-12 weeks after starting therapy and then annually if stable. With the elderly and patients' renal impairment creatine kinase (CK) baseline measurements as they have a higher risk of myopathy.

Managing Inadequate Response or Side Effects

Inadequate response to lipid-lowering therapy may result from several factors that should be systematically evaluated:

- Poor adherence (most common cause)
- Inappropriate drug selection or dosing
- Secondary causes of dyslipidemia (uncontrolled diabetes, hypothyroidism, nephrotic syndrome, cholestasis, medications)
- Genetic factors affecting drug metabolism or severe underlying genetic dyslipidemia
- Lifestyle factors (diet high in saturated fat, weight gain, alcohol consumption)

For statin-associated muscle symptoms (SAMS), which affect approximately 5-10% of patients in clinical practice, a structured approach includes:

1. Temporary discontinuation to determine if symptoms resolve
2. Rule out other causes of muscle symptoms (hypothyroidism, vitamin D deficiency, recent exercise).
3. Rechallenge with the same or different statin at lower dose
4. Consider alternate dosing strategies (every other day, twice weekly) with longer half-life statins (atorvastatin, rosuvastatin)
5. If intolerant to multiple statins, consider non-statin therapies (ezetimibe, PCSK9 inhibitors, bempedoic acid)

Managing side effects is crucial for maintaining long-term adherence to therapy.

When faced with ina[illegible]ise approach is recommended:

1. Assess and address medication adherence through patient education and simplifying regimens when possible
2. Rule out or treat secondary causes of dyslipidemia
3. Optimize lifestyle modifications (diet, exercise, weight management)
4. Consider dose adjustment or switching to a more potent agent within the same class
5. Add complementary therapy with different mechanism of action

What are the common causes of inadequate lipid response to statin therapy?

CASE CONNECTION

BACK TO INTRODUCTION ↑

Thinking back to ML, how do you decide on initial therapy to address his elevated LDL-C and triglycerid[illegible]nd for ML?

Effective management required identifying his cardiovascular risk, setting lipid targets, and choosing therapies with proven benefit, combining first-line agents as appropriate while monitoring for efficacy and side effects. The course of action for ML is decided using the ASCVD-PREVENT risk score calculator. The labs indicate that his LDL-C levels are high (>70 mg/dL) and he has diabetes You identify the high ASCVD risk patient with diabetes and elevated LDL-C (> 20%) and recommend high-intensity statin (atorvastatin or rosuvastatin) therapy plus lifestyle interventions. You justify the statin choice using the ASCVD risk calculator output.

Summary

- Dyslipidemia treatment aims to reduce ASCVD risk by targeting LDL-C and triglycerides, with individualized goals based on cardiovascular risk profile.
- Statins are the first-line therapy for LDL-C reduction, with ezetimibe and PCSK9 inhibitors as complementary options for patients not achieving goals on statin monotherapy.
- Triglyceride management focuses on levels ≥150 mg/dL, with fibrates, prescription omega-3 fatty acids, and lifestyle modifications as primary interventions.
- ASCVD risk assessment using tools like the Pooled Cohort Equations guides therapy decisions, with treatment thresholds based on absolute risk and baseline lipid levels.
- Combination therapy for LDL-C involves maximizing statin dose, adding ezetimibe, and considering PCSK9 inhibitors for high-risk patients not at goal.
- Regular monitori[illegible]articularly statin-associated muscle symptoms, are essential for maintaining effective dyslipidemia treatment.

Review Questions

1. What is the primary role of LDL-C in ASCVD risk?

A. ○ Triglyceride metabolism mediator
B. ● Arterial plaque formation contributor
C. ○ Protective cholesterol form
D. ○ Primary HDL precursor
E. ○ Lipoprotein(a) component

Hide Explanation

The correct answer is (B). LDL-C contributes causally to atherosclerosis by promoting arterial plaque formation, significantly increasing ASCVD risk. (A) Triglycerides contribute but through complex mechanisms and less directly. (C) HDL-C is protective, not causal. (D) VLDL is a precursor but less targeted than LDL-C. (E) Lipoprotein(a) is a risk factor but not the main LDL-C contributor.

2. What does the ASCVD risk calculator primarily assess?

A. ○ Coronary artery stenosis degree
B. ○ Liver function status
C. ● 10-year cardiovascular risk
D. ○ Blood pressure category
E. ○ Triglyceride level assessment

Hide Explanation

(C) is correct because the ASCVD risk calculator estimates a 10-year risk of first ASCVD events like myocardial infarction or stroke. (A) Coronary artery stenosis is assessed by imaging, not the calculator. (B) Liver function is unrelated. (D) Blood pressure classification is a component but not the overall purpose. (E) Triglyceride levels are inputs but not the primary outcome assessed.

3. Which agent lowers LDL-C by inhibiting intestinal cholesterol absorption?

A. ○ Statins
B. ● Ezetimibe
C. ○ PCSK9 inhibitors
D. ○ Fibrates
E. ○ Niacin

Hide Explanation

(B) is correct; Ezetimibe blocks NPC1L1 protein to reduce cholesterol absorption. (A) Statins inhibit HMG-CoA reductase in the liver. (C) PCSK9 inhibitors prevent LDL receptor degradation. (D) Fibrates lower triglycerides via PPAR-alpha activation. (E) Niacin reduces triglycerides by lowering lipolysis, not absorption.

4. When should LDL-C lowering therapy be initiated in a patient with diabetes aged 40-75?

A. ○ Only if LDL-C≥190 mg/dL
B. ○ Start if triglycerides≥500
C. ● Start if LDL-C≥70 mg/dL
D. ○ Only if 10-year risk≥20%
E. ○ Avoid statins due to diabetes

Hide Explanation

(C) is correct; guidelines recommend at least moderate-intensity statin therapy if LDL-C is ≥70 mg/dL i[illegible] is not indicated regardless of LDL-C in this group. (B) Threshold 190 mg/dL applies mainly to primary prevention without diabetes. (D) Therapy at LDL-C <5% risk is only considered if LDL-C ≥190 mg/dL. (E) Statin therapy is not avoided in diabetes despite small increased diabetes risk.

5. What is the recommended next step if LDL-C remains above target despite maximally tolerated statin?

A. ○ Switch to fibrates
B. ● Add ezetimibe
C. ○ Add fibrates
D. ○ Add niacin immediately
E. ○ Stop statins and observe

Hide Explanation

(B) is correct; adding ezetimibe is the next step due to its complementary mechanism. (A) Increasing statin dose beyond tolerated limits is inappropriate. (C) Starting fibrates targets triglycerides, not LDL-C primarily. (D) Starting niacin is limited by lack of outcome benefit. (E) Discontinuation without alternative therapy leads to uncontrolled LDL-C.

(B) is correct; adding ezetimibe is the next step due to its complementary mechanism. (A) Increasing statin dose beyond tolerated limits is inappropriate. (C) Starting fibrates targets triglycerides, not LDL-C primarily. (D) Starting niacin is limited by lack of outcome benefit. (E) Discontinuation without alternative therapy leads to uncontrolled LDL-C.

How would you rate this Brick?

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Explain indications for LDL-C and triglyceride lowering therapy based on patient risk factors~~
02 ☑ ~~Apply treatment guidelines and ASCVD risk assessment to select appropriate lipid-lowering therapies~~
03 ☑ ~~Evaluate patient scenarios to optimize and adjust dyslipidemia treatment strategies~~

Objective Confidence: 3/3